EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. A spherical balloon is being inflated at the rate of 35 cc/min. The rate of increase in the surface area (in cm^2/min.) of the balloon when its diameter is 14 cm, is : (Use $\pi = \frac{22}{7}$)

(A) $\sqrt{10}$ (B) $10\sqrt{10}$ (C) 100 (D) 10

MTN001

2. Let S be a square with sides of length x. If we approximate the change in size of the area of S by $h \cdot \left.\frac{dA}{dx}\right|_{x=x_0}$, when the sides are changed from x_0 to $x_0 + h$, then the absolute value of the error in our approximation, is

(A) h^2 (B) $2hx_0$ (C) x_0^2 (D) h

MTN051

3. The slope of the curve $y = \sin x + \cos^2 x$ is zero at the point, where-

(A) $x = \frac{\pi}{4}$ (B) $x = \frac{\pi}{2}$ (C) $x = \pi$ (D) No where

MTN002

4. The equation of tangent at the point (at^2, at^3) on the curve $ay^2 = x^3$ is-

(A) $3tx - 2y = at^3$ (B) $tx - 3y = at^3$ (C) $3tx + 2y = at^3$ (D) None of these

MTN003

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5. The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to the x-axis, is :-

(A) $y = 0$ (B) $y = 1$ (C) $y = 2$ (D) $y = 3$

MTN052

6. The normal to the curve, $x^2 + 2xy - 3y^2 = 0$, at (1, 1) :

(A) meets the curve again in the third quadrant

(B) meets the curve again in the fourth quadrant

(C) does not meet the curve again

(D) meets the curve again in the second quadrant

MTN053

7. The coordinates of the points on the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, where tangent is inclined an angle $\pi/4$ to the x-axis are -

(A) (a, a) (B) $\left(a\left(\frac{\pi}{2}-1\right), a\right)$ (C) $\left(a\left(\frac{\pi}{2}+1\right), a\right)$ (D) $\left(a, a\left(\frac{\pi}{2}+1\right)\right)$

MTN004

8. Consider the curve represented parametrically by the equation

$x = t^3 - 4t^2 - 3t$ and $y = 2t^2 + 3t - 5$ where $t \in R$.

If H denotes the number of point on the curve where the tangent is horizontal and V the number of point where the tangent is vertical then

(A) $H = 2$ and $V = 1$ (B) $H = 1$ and $V = 2$ (C) $H = 2$ and $V = 2$ (D) $H = 1$ and $V = 1$

MTN054

For More Join: @IITJEE_Advanced

9. The point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi axis OX a line segment half that on the negative semi axis OY then the co-ordinates of the point(s) is/are given by :

(A) $(-1, 9)$ (B) $(3, -15)$ (C) $(1, -3)$ (D) none

MTN055

10. If the tangent at a point P, with parameter t, on the curve $x = 4t^2 + 3, y = 8t^3 - 1, t \in R$, meets the curve again at a point Q, then the coordinates of Q are :

(A) $(t^2 + 3, t^3 - 1)$ (B) $(t^2 + 3, -t^3 - 1)$

(C) $(16t^2 + 3, -64t^3 - 1)$ (D) $(4t^2 + 3, -8t^3 - 1)$

MTN056

11. The curve $x^2 - y^2 = 5$ and $\frac{x^2}{18} + \frac{y^2}{8} = 1$ cut each other at any common point at an angle-

(A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) None of these

MTN006

12. The lines tangent to the curve $y^3 - x^2 + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to -

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

MTN057

13. The angle of intersection between the curves $y^2 = 8x$ and $x^2 = 4y - 12$ at (2, 4) is-

(A) 90° (B) 60° (C) 45° (D) 0°

MTN007

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14. If the curves $y^2 = 6x, 9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is :

(A) $\frac{7}{2}$ (B) 4 (C) $\frac{9}{2}$ (D) 6

MTN058

15. The length of subtangent to the curve $x^2 + xy + y^2 = 7$ at the point (1, –3) is -

(A) 3 (B) 5 (C) 15 (D) 3/5

MTN008

16. At any point of a curve (subtangent) × (subnormal) is equal to the square of the-

(A) slope of the tangent at that point (B) slope of the normal at that point

(C) abscissa of that point (D) ordinate of that point

MTN009

17. For a curve $\frac{(\text{length of normal})^2}{(\text{length of tangent})^2}$ is equal to -

(A) (subnormal)/(subtangent) (B) (subtangent)/(subnormal)

(C) (subtangent × subnormal) (D) constant

MTN059

18. A line L is perpendicular to the curve $y = \frac{x^2}{4} - 2$ at its point P and passes through (10, –1). The coordinates of the point P are

(A) (2, –1) (B) (6, 7) (C) (0, –2) (D) (4, 2)

MTN010

For More Join: @IITJEE_Advanced

19. At the point $P(a, a^n)$ on the graph of $y = x^n$ $(n \in N)$ in the first quadrant a normal is drawn. The normal intersects the y-axis at the point $(0, b)$. If $\lim_{a\to 0} b = \frac{1}{2}$, then n equals

(A) 1 (B) 3 (C) 2 (D) 4

MTN060

20. A circle with centre at (15, –3) is tangent to $y = \frac{x^2}{3}$ at a point in the first quadrant. The radius of the circle is equal to

(A) $5\sqrt{6}$ (B) $8\sqrt{3}$ (C) $9\sqrt{2}$ (D) $6\sqrt{5}$

MTN061

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For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $\frac{x}{a}+\frac{y}{b}=1$ is a tangent to the curve $x = Kt, y = \frac{K}{t}, K > 0$ then:

(A) $a > 0,\ b > 0$ (B) $a > 0,\ b < 0$ (C) $a < 0, b > 0$ (D) $a < 0, b < 0$

MTN029

2. For the curve $C : y = e^{2x}\cos x$, which of the following statement(s) is/are true ?

(A) equation of the tangent where C crosses y-axis is $y = 3x + 1$

(B) equation of the tangent where C crosses y-axis is $y = 2x + 1$

(C) number of points in $\left(-\frac{\pi}{2},\frac{3\pi}{2}\right)$ where tangent on the curve C is parallel to x-axis is 4.

(D) number of points in $\left(-\frac{\pi}{2},\frac{3\pi}{2}\right)$ where tangent on the curve C is parallel to x-axis is 2.

MTN030

3. The equation of normal to curve $\left(\frac{x}{a}\right)^n+\left(\frac{y}{b}\right)^n=2$, where $n \in N$, at the point with abscissa equal to 'a' can be -

(A) $ax + by = a^2 - b^2$ (B) $ax + by = a^2 + b^2$ (C) $ax - by = a^2 - b^2$ (D) $bx + ay = a^2 - b^2$

MTN033

4. Given the curve $\left(\frac{x^2-y^2}{xy}\right)^3=\frac{512}{27}$, then

(A) the slope of normal drawn to the curve at (3,1) is $\frac{1}{3}$

(B) the value of $\left(2\frac{dy}{dx}-5\frac{d^2y}{dx^2}\right)_{(3,1)}=\frac{2}{3}$

(C) the slope of normal drawn to the curve at (3,1) is –3.

(D) the equation of tangent at (3,1) to given curve is $3y + x = 0$.

MTN032

5. Given that $g(x)$ is a non constant linear function defined on R-

(A) $y = g(x)$ and $y = g^{-1}(x)$ are orthogonal

(B) $y = g(x)$ and $y = g^{-1}(-x)$ are orthogonal

(C) $y = g(-x)$ and $y = g^{-1}(x)$ are orthogonal

(D) $y = g(-x)$ and $y = g^{-1}(-x)$ are orthogonal

MTN062

6. If the curve $y^2 = 4ax, a > 0$ cuts the curve $xy = b, b > 0$ at right angle, then -

(A) $a=\frac{1}{\sqrt{2}},b=\sqrt{2}$ (B) $a=\frac{1}{2},b=\sqrt{2}$ (C) $a=\frac{1}{4},b=\frac{1}{2\sqrt{2}}$ (D) $a=\frac{1}{4},b=\sqrt{2}$

MTN063

7. For the curve $x = t^2 + 3t - 8, y = 2t^2 - 2t - 5$, at point (2, –1)

(A) length of subtangent is 7/6 (B) slope of tangent = 6/7

(C) length of tangent = $\sqrt{(85)}/6$ (D) none of these

MTN031

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For More Join: @IITJEE_Advanced

8. If curves $f(x) = x^2 + ax + b$ and $g(x) = cx - x^2$ touches each other at point $(1, 0)$, then

(A) $a = -3$ (B) $b = 2$ (C) $c = 1$ (D) $abc = -6$

MTN034

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 9 to 10

Water is flowing out at the rate of 6 m^3/min from a reservoir shaped like a hemispherical bowl of radius $R = 13m$. The volume of water in the hemispherical bowl is given by $V = \frac{\pi}{3} \cdot y^2(3R - y)$ when the water is y meter deep. Find

9. At what rate is the water level changing when the water is $8m$ deep.

(A) $\frac{-1}{24\pi} m/min.$ (B) $\frac{-5}{288\pi} m/min.$ (C) $\frac{-1}{12\pi} m/min.$ (D) $\frac{-5}{144\pi} m/min.$

MTN038

10. At what rate is the radius of the water surface changing when the water is $8m$ deep.

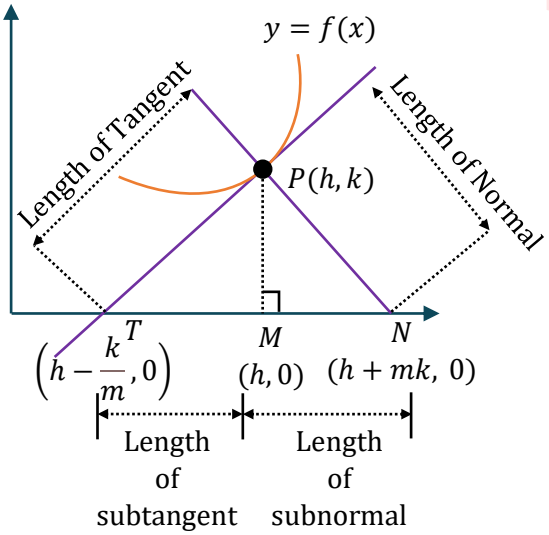
(A) $\frac{-1}{24\pi} m/min.$ (B) $\frac{-5}{288\pi} m/min.$ (C) $\frac{-1}{12\pi} m/min.$ (D) $\frac{-5}{144\pi} m/min.$

MTN039

Paragraph for Question No. 11 to 13

Let $P(h, k)$ be any point on curve $y = f(x)$. Let tangent drawn at point P meets x-axis at T & normal at point P meets x-axis at N (as shown in figure) and $m = \left.\frac{dy}{dx}\right]_{(h,k)}$ = slope of tangent.

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(i) Length of Tangent = $PT = |k|\sqrt{1 + \frac{1}{m^2}}$

(ii) Length of Normal = $PN = |k|\sqrt{1 + m^2}$

(iii) Length of subtangent = Projection of segment PT on x-axis = $TM = \left|\frac{k}{m}\right|$

(iv) Length of subnormal = projection of line segment PN on x axis = $MN = |km|$

For More Join: @IITJEE_Advanced

11. The product of length of tangent and length of normal for the curve $y = x^3 + 3x^2 + 4x - 1$ at point $x = 0$ is

(A) $\frac{17}{4}$ (B) $\frac{\sqrt{15}}{4}$ (C) 17 (D) $\frac{4}{\sqrt{17}}$

MTN035

12. Value of 'p' such that the length of the subtangent and subnormal is equal for the curve $y = e^{px} + px$ at the point (0, 1)

(A) ±1 (B) ±2 (C) $\pm\frac{1}{2}$ (D) $\pm\frac{1}{4}$

MTN036

13. The length of subnormal to $x = \sqrt{2}\cos t$, $y = -3\sin t$ at $t = \frac{-\pi}{4}$.

(A) $\frac{2}{9}$ (B) 1 (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

MTN037

MATCHING LIST TYPE QUESTION

14. Consider the curve $y = e^{2x^2} + 2x$, then match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	Length of tangent at $x = 0$ is	(1)	$\sqrt{5}$
(Q)	Length of normal at $x = 0$ is	(2)	$\frac{1}{2}$
(R)	Length of subtangent at $x = 0$ is	(3)	$\frac{\sqrt{5}}{2}$
(S)	Length of subnormal at $x = 0$ is	(4)	2

(A) P → 1; Q → 3; R → 2; S → 4 (B) P → 4; Q → 3; R → 1; S → 2

(C) P → 3; Q → 1; R → 2; S → 4 (D) P → 3; Q → 1; R → 4; S → 2

MTN040

MATRIX MATCH TYPE QUESTION

15.

	Column-I		**Column-II**
(A)	The slope of the curve $2y^2 = ax^2 + b$ at (1, –1) is –1, then	(p)	$a - b = 2$
(B)	If (a, b) be the point on the curve $9y^2 = x^3$ where normal to the curve makes equal intercepts with the axes, then	(q)	$a - b = 7/2$
(C)	If the tangent at a point (1, 2) on the curve $y = ax^2 + bx + \frac{7}{2}$ be parallel to the normal at (–2, 2) on the curve $y = x^2 + 6x + 10$, then	(r)	$a - b = 4/3$
(D)	If the tangent to the curve $xy + ax + by = 0$ at (1, 1) is inclined at an angle $\tan^{-1} 2$ with x-axis, then	(s)	$a - b = 3$

MTN041

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For More Join: @IITJEE_Advanced

EXERCISE (S)

1. Water is being poured on to a cylindrical vessel at the rate of 1 m^3/min and the vessel has a circular base of radius 3 m. If the rate at which the level of water is rising in the vessel is $\frac{1}{k\pi}$ m/min, then the value of 'k' is

MTN011

2. An inverted cone has a depth of 10 cm & a base of radius 5 cm. Water is poured into it at the rate of 1.5 cm^3/min. If the rate at which level of water in the cone is rising, when the depth of water is 4 cm is $\frac{3}{k\pi}$ cm/min, then the value of 'k' is

MTN064

3. Water is dripping out from a conical funnel of semi vertical angle $\pi/4$, at the uniform rate of 2 cm^3/sec through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, then the rate of decrease of the slant height of the water is given by $\frac{\sqrt{2}}{k\pi}$ cm/sec, then the value of 'k' is

MTN042

4. A water tank has the shape of a right circular cone with its vertex down. Its altitude is 10 cm and the radius of the base is 15 cm. Water leaks out of the bottom at a constant rate of $1cu.cm/sec$. Water is poured into the tank at a constant rate of C $cu.cm/sec$. Compute C so that the water level will be rising at the rate of 4 cm/sec at the instant when the water is 2 cm deep. If C is of the form $p + q\pi$, then the value of $p + q$ is

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MTN065

5. Sand is pouring from a pipe at the rate of 12 cc/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always $1/6^{th}$ of the radius of the base. If the height of the sand cone increase at a rate $\frac{1}{p\pi}$ when the height is 4 cm, then the value of 'p' is

MTN066

6. A circular ink blot grows at the rate of 2 cm^2 per second. Then the rate at which the radius is increasing after $2\frac{6}{11}$ seconds is (Use $\pi = \frac{22}{7}$)

MTN043

7. The tangent to $y = ax^2 + bx + \frac{7}{2}$ at (1, 2) is parallel to the normal at the point (–2, 2) on the curve $y = x^2 + 6x + 10$. Then value of $a - 2b$ is

MTN012

8. The chord of the parabola $y = -a^2x^2 + 5ax - 4$ touches the curve $y = \frac{1}{1-x}$ at the point $x = 2$ and is bisected by that point, then the value of 'a' is

MTN013

For More Join: @IITJEE_Advanced

9. A variable ΔABC in the xy plane has its orthocentre at vertex 'B' , a fixed vertex 'A' at the origin and the third vertex 'C' restricted to lie on the parabola $y = 1 + \frac{7x^2}{36}$. The point B starts at the point $(0, 1)$ at time $t = 0$ and moves upward along the y axis at a constant velocity of 2 cm/sec, then how fast is the area of the triangle increasing when $t = \frac{7}{2} sec$.

MTN044

10. There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = -8/x$ where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves at these points respectively then the value of $(p + q)$ is

MTN045

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. The tangent to the curve, $y = xe^{x^2}$ passing through the point $(1, e)$ also passes through the point : **[JEE (Main) 2019]**

(1) $\left(\frac{4}{3}, 2e\right)$ (2) $(2, 3e)$ (3) $\left(\frac{5}{3}, 2e\right)$ (4) $(3, 6e)$

MTN014

2. A $2m$ ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 $cm/sec.$, then the rate (in $cm/sec.$) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is $1m$ above the ground is : **[JEE (Main) 2019]**

(1) $25\sqrt{3}$ (2) 25 (3) $\frac{25}{\sqrt{3}}$ (4) $\frac{25}{3}$

MTN015

3. The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point $(2, 2)$ is **[JEE (Main) 2020]**

(1) $4\sqrt{2}$ (2) $2\sqrt{2}$ (3) 2 (4) $\sqrt{2}$

MTN016

4. Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0, \frac{3}{2}\right)$. If m is the slope of the tangent at P to the curve, then $|m|$ is equal to **[JEE (Main) 2020]**

MTN017

5. Let $P(h, k)$ be a point on the curve $y = x^2 + 7x + 2$, nearest to the line, $y = 3x - 3$. Then the equation of the normal to the curve at P is : **[JEE (Main) 2020]**

(1) $x + 3y - 62 = 0$ (2) $x - 3y - 11 = 0$ (3) $x - 3y + 22 = 0$ (4) $x + 3y + 26 = 0$

MTN018

6. The equation of the normal to the curve $y = (1 + x)^{2y} + \cos^2(\sin^{-1}x)$ at $x = 0$ is :**[JEE (Main) 2020]**

(1) $y = 4x + 2$ (2) $x + 4y = 8$ (3) $y + 4x = 2$ (4) $2y + x = 4$

MTN019

7. Which of the following points lies on the tangent to the curve $x^4e^y + 2\sqrt{y+1} = 3$ at the point $(1, 0)$?

[JEE (Main) 2020]

(1) $(2, 2)$ (2) $(-2, 6)$ (3) $(-2, 4)$ (4) $(2, 6)$

MTN020

8. If the tangent to the curve, $y = f(x) = x \log_e x$, $(x > 0)$ at a point $(c, f(c))$ is parallel to the line - segement joining the points $(1, 0)$ and (e, e), then c is equal to : **[JEE (Main) 2021]**

(1) $\frac{1}{e-1}$ (2) $e^{\left(\frac{1}{1-e}\right)}$ (3) $e^{\left(\frac{1}{e-1}\right)}$ (4) $\frac{e-1}{e}$

MTN021

For More Join: @IITJEE_Advanced

9. An angle of intersection of the curves, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $x^2+y^2=ab$, $a > b$, is : **[JEE (Main) 2021]**

(1) $\tan^{-1}\left(\frac{a+b}{\sqrt{ab}}\right)$ (2) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ (3) $\tan^{-1}\left(\frac{a-b}{\sqrt{ab}}\right)$ (4) $\tan^{-1}\left(2\sqrt{ab}\right)$

MTN022

10. If the curve $y = ax^2 + bx + c$, $x \in R$, passes through the point (1, 2) and the tangent line to this curve at origin is $y = x$, then the possible values of a, b, c are : **[JEE (Main) 2021]**

(1) $a = \frac{1}{2}$, $b = \frac{1}{2}$, $c = 1$ (2) $a = 1$, $b = 0$, $c = 1$

(3) $a = 1$, $b = 1$, $c = 0$ (4) $a = -1$, $b = 1$, $c = 1$

MTN067

11. If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a , b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point $(2, -1)$, then $|2a + 9b|$ is equal to ____. **[JEE (Main) 2022]**

MTN068

12. If the tangent to the curve $y = x^3$ at the point $P(t, t^3)$ meets the curve again at Q, then the ordinate of the point which divides PQ internally in the ratio 1 : 2 is : **[JEE (Main) 2022]**

(1) $-2t^3$ (2) 0 (3) $-t^3$ (4) $2t^3$

MTN023

13. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is : **[JEE (Main) 2022]**

(1) 9 (2) 10 (3) 11 (4) 12

MTN024

14. The tangent at the point (x_1 , y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve : **[JEE (Main) 2022]**

(1) $x^2+\frac{y^2}{81}=2$ (2) $\frac{y^2}{9}-x^2=8$ (3) $y=4x^2+5$ (4) $\frac{x}{3}-y^2=2$

MTN025

15. Water is being filled at the rate of 1 cm^3 / sec in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is **[JEE (Main) 2022]**

(1) 5 (2) $\frac{\sqrt{21}}{5}$ (3) $\frac{\sqrt{26}}{5}$ (4) $\frac{\sqrt{26}}{10}$

MTN026

16. If the angle made by the tangent at the point (x_0, y_0) on the curve $x = 12(t + \sin t \cos t)$, $y=12(1+\sin t)^2, 0<t<\frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to **[JEE (Main) 2022]**

(1) $6\left(3+2\sqrt{2}\right)$ (2) $3\left(7+4\sqrt{3}\right)$ (3) 27 (4) 48

MTN027

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17. A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____. **[JEE (Main) 2022]**

MTN028

18. If the equation of the normal to the curve $y=\frac{x-a}{(x+b)(x-2)}$ at the point (1, –3) is $x - 4y = 13$, then the value of $a + b$ is equal to. **[JEE (Main) 2023]**

MTN069

19. The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is : **[JEE (Main) 2023]**

(1) 2 (2) 3 (3) 4 (4) 0

MTN070

20. Let quadratic curve passing through the point (–1, 0) and touching the line $y = x$ at (1, 1) be $y = f(x)$. Then the x-intercept of the normal to the curve at the point $(\alpha, \alpha + 1)$ in the first quadrant is.

[JEE (Main) 2023]

MTN071

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EXERCISE - JEE (Advanced) PYQ

1. Find the equation of the straight line which is tangent at one point and normal at another point of the curve, $x = 3t^2$, $y = 2t^3$. **[JEE (Advanced) 2000]**

MTN046

2. If the normal to the curve , $y = f(x)$ at the point (3, 4) makes an angle $\frac{3\pi}{4}$ with the positive x–axis. Then $f'(3) =$ **[JEE (Advanced) 2000]**

(A) – 1 (B) $-\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 1

MTN047

3. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical, is(are) **[JEE (Advanced) 2002]**

(A) $\left(\pm\frac{4}{\sqrt{3}}, -2\right)$ (B) $\left(\pm\sqrt{\frac{11}{3}}, 1\right)$ (C) (0, 0) (D) $\left(\pm\frac{4}{\sqrt{3}}, 2\right)$

MTN048

4. Tangent to the curve $y = x^2 + 6$ at a point $P(1, 7)$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q. Then the coordinates of Q are **[JEE (Advanced) 2005]**

(A) (– 6, –11) (B) (–9, –13) (C) (– 10, – 15) (D) (–6, –7)

MTN049

Rankers Academy JEE

5. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$ **[JEE (Advanced) 2007]**

(A) on the left of $x = c$ (B) on the right of $x = c$

(C) at no point (D) at all points

MTN050

6. The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point (1,3) is **[JEE (Advanced) 2014]**

MTN120

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	B	A	D	B	C	B	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	D	C	C	D	A	D	C	D

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,D	B,D	A,C	B,C	B,C	B,C	A,B,C	A,B,C,D	A	B
Que.	11	12	13	14	15					
Ans.	A	C	D	C	A → p; B → r; C → q; D → s					

EXERCISE (S)

1.	9.00	**2.**	8.00	**3.**	4.00	**4.**	37.00	**5.**	48.00
6.	0.25	**7.**	6.00	**8.**	1.00	**9.**	9.42 or 9.43	**10.**	20.00

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	2	4.00	4	2	2	3	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	195	1	1	4	3	3	5	4	3	11

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	
Ans.	$\sqrt{2}x - y - 2\sqrt{2} = 0$	D	D	D	A	8	

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Important Notes

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